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a) Lets use the series from Q2c) for this.
Recall $X_t = Z_0 S_t + Z_1 S_{t-1}$.
We have already proven it is weakly stationary.

Since the seasonality has a period of 2, then we can conclude that $X_t = \begin{cases} Z_0 S_t & \text{when } t \text{ is even} \\ Z_1 S_{t-1} & \text{when } t \text{ is odd} \end{cases}$

As such, we need only assume that Z_0 and Z_1 have a different distribution for their random variables (e.g. $Z_0 \sim \mathcal{N}(0,1)$ and $Z_1 \in \{-1, 1\}$ w/ 0.5 prob. each).
If this is the case, then it is easy to see that X_t and X_{t+1} will not have the same joint distribution.
Even t will have $P(Z_0 = 1) = 0$ (property of normal distributions), and $P(Z_1 = 1) = 0.5$.
These two are different and immediately show that $\nexists t$, $X_t \neq X_{t+k}$ for $k \in \mathbb{N}$, thus violating the definition of strong stationarity.